

PhD Concept Note

AI for Colonial Heritage:
NLP and Spatial Machine Learning for Pre-Colonial
Indonesian Settlement Reconstruction from VOC Archives

Prepared for Dr. Houda Lamqaddam and Dr. Delfina Martinez Pandiani
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(with Prof. Tobias Blanke as promoter)

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1 Summary

This concept note outlines a PhD project that applies NLP and spatial machine learning to Dutch East India Company (VOC) and late-colonial Dutch archives (1602–1942) in order to reconstruct pre-colonial Indonesian settlement geography. The project treats systematic bias in the archaeological and textual record — colonial, volcanic, historiographic — as the object of study, rather than as noise to be averaged out. It produces three peer-reviewed publications, a reusable colonial settlement gazetteer, and an open historical Dutch NER resource. The project fits HAICu’s mandate of AI for unlocking colonial heritage collections and extends the AI, Culture & Society research line into a domain where $\geq 99.94\%$ of the expected pre-1600 archaeological signal is physically missing (E133) and must be reconstructed from text.

2 The Archaeological Gap

Conventional archaeological survey in volcanic Indonesia recovers almost none of the expected pre-colonial record. A burial model calibrated against 363 site depths (E075, Pearson $r = 0.951$) and cross-referenced with 51 eruption-site pairs (E083) yields a median burial depth of 2.5 m and a mean sedimentation rate of 4.4 mm/yr. Over a 1,000-year pre-colonial window this places typical settlements well below the ~ 2 m threshold of standard archaeological survey. A cascade model (E133) estimates that $\geq 99.94\%$ of the expected pre-1600 settlement signal in volcanic Java is archaeologically invisible.

The Philippines, at comparable pre-colonial population density but with non-volcanic (karst) geology, has 4,000+ pre-colonial sites. Volcanic Java has zero confirmed open-air pre-400 CE sites. Monte Carlo population estimation (E196) places 1–2 million people in Java at 400 CE, implying a minimum of 694 expected sites. The taphonomic suppression factor is $\geq 694\times$.

3 Why Colonial Dutch Archives

If the physical record is inaccessible, reconstruction must come from textual, epigraphic, or remote-sensing sources. The highest-volume textual source is the VOC archive and its colonial-era successors.

A pilot rule-based extraction from 16 volumes of the colonial archaeological service reports (OV, 1912–1929) alone yielded 22,162 structured mentions — 6,932 named sites, 4,933 administrative locations, 9,238 material descriptions (E091). A separate Delpher extraction (E141) recovered 1,768 records from 46 queries. GLOBALISE (Huygens Institute / VU Amsterdam) is producing structured text from 5M+ handwritten VOC pages. The full corpus — spanning 340 years of dagregisters, scientific journals, technical reports and newspapers — remains largely unmined.

Colonial infrastructure projects (railways, canals, wells) produced thousands of incidental archaeological observations recorded in newspapers and technical reports — observations made *without* archaeological selection bias, a rare property in the source material.

4 Three AI Challenges

Challenge 1 — NER and IE for historical Dutch with code-switching.

Fine-tuned Transformer NER (ArcheoBERTje / BERTje base) on 17th–19th century Dutch with orthographic variation, OCR noise, Dutch–Malay code-switching, and Sanskrit-derived toponyms. Entity types: LOCATION, DEPTH, MATERIAL, TEMPORAL, FIND_EVENT. Preprocessing combines rule-based and neural spelling normalisation with synthetic OCR noise augmentation.

Challenge 2 — Historical place-name disambiguation.

Three-tier pipeline: exact match against a historical gazetteer (World Historical Gazetteer, HISGIS); phonological fuzzy matching for orthographic variants; contextual disambiguation via administrative hierarchy (residency → regency → district). Each record carries a calibrated spatial uncertainty radius.

Challenge 3 — Integration of uncertain textual and spatial evidence.

Bayesian integration of NLP-extracted settlement locations with a prior probability surface from the VOLCARCH sedimentation and visibility model. Each record carries textual-extraction confidence, geocoding uncertainty, and physical-prior consistency. Output: a probabilistic settlement map with quantified, legible uncertainty.

5 Alignment with UvA — AI, Culture & Society

Three concrete connection points with the group’s current work:

1. **Historical structured-data extraction (Lamqaddam).** Recent work applying OCR and LLMs to convert digitised 19th-century exhibition catalogues into structured data maps directly onto the technical core of this project. The VOC/Delpher pipeline is the same problem class at archive scale, in a different linguistic register, and with a spatial validation target.
2. **Ethical and explainable AI for cultural heritage (Martinez Pandiani).** This project treats systematic bias — colonial, taphonomic, historiographic — as the object of analysis rather than a source of error to be minimised. The central question — *whose knowledge gets preserved, and how computational methods can expose rather than reproduce those erasures* — is a heritage-AI ethics question in the operational sense. Explainability is load-bearing here: downstream users (archaeologists, historians, Indonesian heritage institutions) will act on uncertain predictions, so uncertainty must be legible, not hidden behind a point estimate.
3. **HAICu — AI for unlocking colonial heritage collections (Blanke, promoter).** Indonesia’s colonial heritage is among the largest and least computationally explored in the world: the VOC operated across the archipelago for nearly 200 years. The project fits HAICu’s scope as an applied case with an unusual extension into spatial modelling.

6 What Makes This Unusual: Physical Validation

Historical NLP is typically evaluated against other textual annotations. This project has access to a non-textual ground truth: a calibrated geophysical sedimentation model. NLP-extracted depth measurements from colonial reports provide an independent test — if the model is correct, colonial finds near volcanoes should be reported at greater depths than finds far from them. A pilot (E197, 33 colonial depth records from Delpher, 1870–1941) gave Wilcoxon $p = 0.131$ (consistent with model predictions; $n = 33$ is under-powered for the depth-distribution test alone) and $5.8\times$ enrichment of finds near VOLCARCH-predicted target zones ($p < 0.00001$). To my knowledge this is an unusual validation paradigm for historical text mining.

7 Preliminary Results

- **E091.** Rule-based Dutch NLP on 16 OV volumes (259K lines). 22,162 structured mentions; 94.2% recall against manual gold standard.
- **E141.** Delpher extraction via KB SRU API. 1,768 records, 46 queries, 165 geocoded, 9 archaeological depth measurements.
- **E197.** Cross-century depth validation. 33 colonial records vs. sedimentation model; Wilcoxon $p = 0.131$; $5.8\times$ enrichment at predicted target zones.
- **Six VOLCARCH papers under review** (EGQSJ, JCAA, Antiquity, Oceanic Linguistics, Archipel, Archeologia e Calcolatori). Preprints: Zenodo DOI 10.5281/zenodo.19081502 (taphonomic framework, E&G submission); arXiv:2604.00023 (phonological ML, Oceanic Linguistics submission).

8 Thesis Structure (Cumulative)

Paper	Target venue	Content
1	ACL / EMNLP / LREC	NER for historical colonial Dutch: model, evaluation, error analysis
2	TACL / DSH	Temporal IE and place-name disambiguation; colonial gazetteer; spatial uncertainty quantification
3	ACL / EACL	Bayesian integration of NLP extractions with the VOLCARCH prior; explainable uncertainty propagation for heritage applications

9 Funding and Timeline

Multiple funding pathways are being explored. Understanding that HAICu and DEEP CULTURE are fully allocated, the candidate seeks (a) guidance on upcoming EU grant calls (NWO, ERC, MSCA Doctoral Networks) where a VOC-archive + spatial-ML project might fit, and, as a parallel option, (b) Indonesia’s **BPI Dosen** scholarship — a fully-funded programme for permanent Indonesian lecturers (tuition and living allowance, up to 4 years). A Letter of Acceptance from UvA would materially strengthen a BPI Dosen application. Flexible on start date: October 2027 or October 2028.

10 Candidate Profile (short)

- M.Sc. Computer Science, Universitas Indonesia (2016). Lecturer, UBHINUS, since 2010. Courses: Machine Learning, NLP, Computer Vision.

- NLP portfolio: Indonesian NLP survey (arXiv:2304.02746), morphological analysis library (ModernKataKupas, PyPI), low-resource dataset creation.
- VOLCARCH: 207 tracked experiments, 6 papers under review, two preprints (Zenodo 10.5281/zenodo.19081502; arXiv:2604.00023). Public repository: <https://github.com/neimasilk/volcarch-repo>.
- Compute: 4 workstations (Intel Core i9, NVIDIA RTX 4080 each), 128 GB RAM aggregate (UBHINUS, Malang).

A full CV is attached separately.